

BOARDNET

Unified Higher Secondary Certificate Examination Communication Network

TECHNICAL DESIGN & IMPLEMENTATION REPORT

Course: Computer Networks (CSE 421)

Base Network Address: 16.65.0.0/16 (Derived from Student ID ending in 1665)

Date: August 31, 2026



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1. Executive Summary

The modernization of communication systems within the educational sector is vital to ensure security, interoperability, and high availability during national examinations. The Ministry of Education of Bangladesh has commissioned the design of BoardNet, a unified digital network infrastructure connecting six regional Higher Secondary Certificate (HSC) education boards: Dhaka (DHK), Rajshahi (RAJ), Chattogram (CTG), Barishal (BAR), Sylhet (SYL), and Khulna (KHU). These boards coordinate examination schedules, manage sensitive student and question data, process examination results, and maintain continuous administrative channels.

BoardNet implements a sophisticated hybrid topology in Cisco Packet Tracer. Dhaka Board (DHK) functions as the central administrative hub, hosting the primary DNS server, web server, and a multi-scope DHCP server. To satisfy physical and dynamic connectivity requirements, the central boards (DHK, SYL, RAJ) are interconnected using a dynamic routing domain running Routing Information Protocol Version 2 (RIPv2). The regional branches (KHU, BAR) utilize point-to-point administrative WAN links with static and floating static routes to achieve recursive path redundancy. The isolated Chattogram Board (CTG) acts as a stub network, connected exclusively to DHK via a serial point-to-point link and utilizing default routing. This design guarantees seamless dynamic IP assignment, robust centralized domain name resolution, unified web services, reliable email correspondence, and automated failover paths.

2. VLSM Design and IP Address Allocation

To distribute IP addresses efficiently and eliminate subnet wastage, a Variable Length Subnet Masking (VLSM) scheme was designed. The subnetting structure is based on a primary network block of 16.65.0.0/16, derived from the first group member's student ID ending in 1665. The address allocation process prioritizes subnets with the largest host requirements to prevent overlap and optimize continuous blocks.

2.1 Sizing Requirements and Logical Partitioning

Each board requires a dedicated Local Area Network (LAN) supporting various dynamic end-user hosts, printers, e-mail servers, and local web or DHCP servers. The table below outlines the host requirements, calculated host bits needed, total allocatable addresses, CIDR block prefixes, and the resulting subnet sizes.

Board / Unit	Est. Host Req.	Host Bits Needed	Usable IPs	Prefix / Mask	Allocated Size
Dhaka (DHK)	300	9 bits	510	/23 (255.255.254.0)	512 IPs
Chattogram (CTG)	200	8 bits	254	/24 (255.255.255.0)	256 IPs

Board / Unit	Est. Host Req.	Host Bits Needed	Usable IPs	Prefix / Mask	Allocated Size
Rajshahi (RAJ)	160	8 bits	254	/24 (255.255.255.0)	256 IPs
Sylhet (SYL)	150	8 bits	254	/24 (255.255.255.0)	256 IPs
Barishal (BAR)	120	7 bits	126	/25 (255.255.255.128)	128 IPs
Khulna (KHU)	120	7 bits	126	/25 (255.255.255.128)	128 IPs

2.2 Allocated LAN Subnets and Gateway Assignments

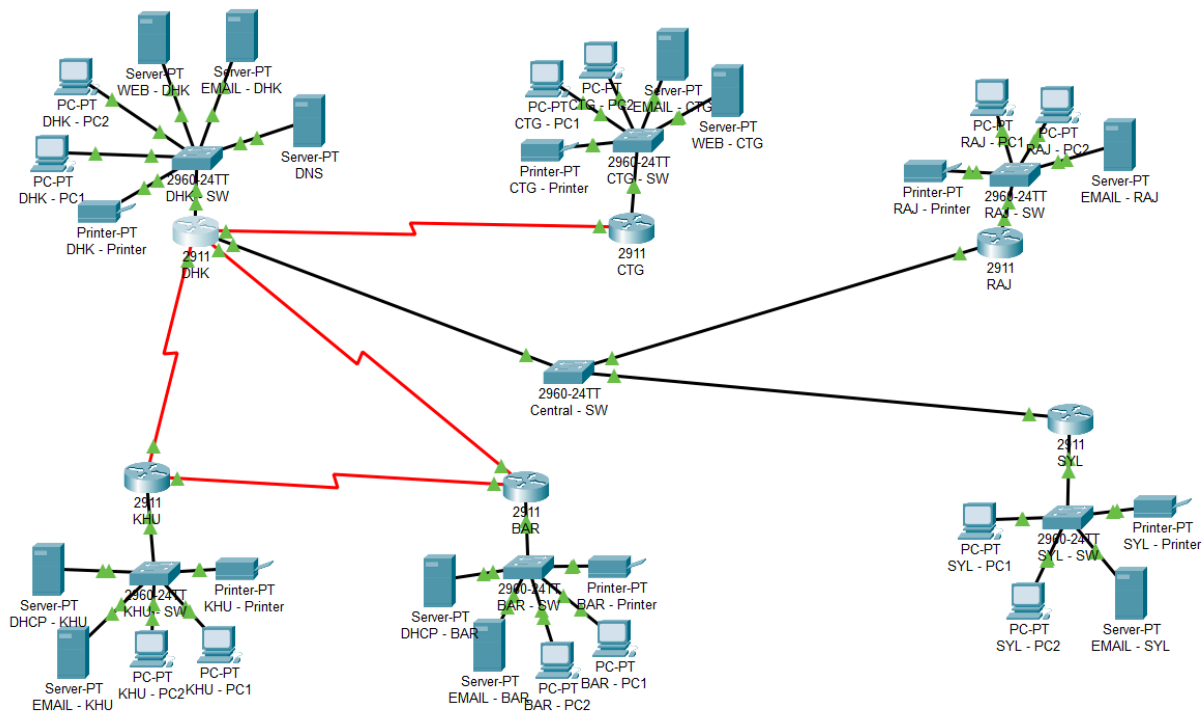
The following table details the actual subnets assigned to each board's LAN, including the usable IP address ranges, broadcast addresses, and the default gateway IP mapped to each router's LAN interface (GigabitEthernet0/0).

Board	Network Address	Subnet Mask	Usable Range	Broadcast	Default Gateway
Dhaka (DHK)	16.65.0.0/23	255.255.254.0	16.65.0.1 - 16.65.1.254	16.65.1.255	16.65.0.1
Chattogram (CTG)	16.65.2.0/24	255.255.255.0	16.65.2.1 - 16.65.2.254	16.65.2.255	16.65.2.1
Rajshahi (RAJ)	16.65.3.0/24	255.255.255.0	16.65.3.1 - 16.65.3.254	16.65.3.255	16.65.3.1
Sylhet (SYL)	16.65.4.0/24	255.255.255.0	16.65.4.1 - 16.65.4.254	16.65.4.255	16.65.4.1
Barishal (BAR)	16.65.5.0/25	255.255.255.128	16.65.5.1 - 16.65.5.126	16.65.5.127	16.65.5.1
Khulna (KHU)	16.65.5.128/25	255.255.255.128	16.65.5.129 - 16.65.5.254	16.65.5.255	16.65.5.129

2.3 Infrastructure WAN Link Subnets

Infrastructure and inter-router links require point-to-point subnets. The Central switch connecting DHK, SYL, and RAJ utilizes a /29 segment to accommodate three active routers on a single broadcast domain, while serial WAN links utilize highly conservative /30 blocks.

Link Segment	Network Address	Subnet Mask	CIDR Block	Configured Router Interfaces
Central Switch Segment	16.65.6.0	255.255.255.248	/29	.1 DHK Gi0/1, .2 SYL Gi0/1, .3 RAJ Gi0/1
Dhaka ↔ Chattogram Serial	16.65.6.8	255.255.255.252	/30	.9 DHK Se0/0/0, .10 CTG Se0/0/0
Dhaka ↔ Khulna Serial	16.65.6.12	255.255.255.252	/30	.13 DHK Se0/0/1, .14 KHU Se0/0/0
Khulna ↔ Barishal Serial	16.65.6.16	255.255.255.252	/30	.17 KHU Se0/0/1, .18 BAR Se0/0/0
Barishal ↔ Dhaka Serial	16.65.6.20	255.255.255.252	/30	.21 BAR Se0/0/1, .22 DHK Se0/1/0



3. Complete Network IP Addressing Table

A rigorous, unified IP addressing master table was compiled for all 46 active endpoints and interfaces across the 6 regional education boards. The table lists every router interface, server, printer, and client PC, showing their static or dynamic IP, subnet mask, default gateway, and configuration method.

Unit	Device Name	Interface	IP Address	Subnet Mask	Default Gateway	Assignment
Dhaka Board	DHK Router	Gi0/0	16.65.0.1	255.255.254.0	—	Manual
Dhaka Board	DHK Router	Gi0/1	16.65.6.1	255.255.255.248	—	Manual
Dhaka Board	DHK Router	Se0/0/0	16.65.6.9	255.255.255.252	—	Manual
Dhaka Board	DHK Router	Se0/0/1	16.65.6.13	255.255.255.252	—	Manual
Dhaka Board	DHK Router	Se0/1/0	16.65.6.22	255.255.255.252	—	Manual
Dhaka Board	DNS Server	Fa0	16.65.0.2	255.255.254.0	16.65.0.1	Manual
Dhaka Board	EMAIL - DHK	Fa0	16.65.0.3	255.255.254.0	16.65.0.1	Manual
Dhaka Board	WEB - DHK	Fa0	16.65.0.4	255.255.254.0	16.65.0.1	Manual
Dhaka Board	DHK - Printer	Fa0	16.65.0.5	255.255.254.0	16.65.0.1	Manual
Dhaka Board	DHK - PC1	Fa0	16.65.0.11	255.255.254.0	16.65.0.1	DHCP (DHK)
Dhaka Board	DHK - PC2	Fa0	16.65.0.12	255.255.254.0	16.65.0.1	DHCP (DHK)
Chattogram	CTG Router	Gi0/0	16.65.2.1	255.255.255.0	—	Manual
Chattogram	CTG Router	Se0/0/0	16.65.6.10	255.255.255.252	—	Manual
Chattogram	EMAIL - CTG	Fa0	16.65.2.2	255.255.255.0	16.65.2.1	Manual
Chattogram	WEB - CTG	Fa0	16.65.2.3	255.255.255.0	16.65.2.1	Manual

Unit	Device Name	Interface	IP Address	Subnet Mask	Default Gateway	Assignment
Chattogram	CTG - Printer	Fa0	16.65.2.4	255.255.255.0	16.65.2.1	Manual
Chattogram	CTG - PC1	Fa0	16.65.2.11	255.255.255.0	16.65.2.1	DHCP (DHK Relay)
Chattogram	CTG - PC2	Fa0	16.65.2.12	255.255.255.0	16.65.2.1	DHCP (DHK Relay)
Rajshahi	RAJ Router	Gi0/0	16.65.3.1	255.255.255.0	—	Manual
Rajshahi	RAJ Router	Gi0/1	16.65.6.3	255.255.255.248	—	Manual
Rajshahi	EMAIL - RAJ	Fa0	16.65.3.2	255.255.255.0	16.65.3.1	Manual
Rajshahi	RAJ - Printer	Fa0	16.65.3.3	255.255.255.0	16.65.3.1	Manual
Rajshahi	RAJ - PC1	Fa0	16.65.3.11	255.255.255.0	16.65.3.1	DHCP (DHK Relay)
Rajshahi	RAJ - PC2	Fa0	16.65.3.12	255.255.255.0	16.65.3.1	DHCP (DHK Relay)
Sylhet	SYL Router	Gi0/0	16.65.4.1	255.255.255.0	—	Manual
Sylhet	SYL Router	Gi0/1	16.65.6.2	255.255.255.248	—	Manual
Sylhet	EMAIL - SYL	Fa0	16.65.4.2	255.255.255.0	16.65.4.1	Manual
Sylhet	SYL - Printer	Fa0	16.65.4.3	255.255.255.0	16.65.4.1	Manual
Sylhet	SYL - PC1	Fa0	16.65.4.11	255.255.255.0	16.65.4.1	DHCP (DHK Relay)
Sylhet	SYL - PC2	Fa0	16.65.4.12	255.255.255.0	16.65.4.1	DHCP (DHK Relay)
Barishal	BAR Router	Gi0/0	16.65.5.1	255.255.255.128	—	Manual
Barishal	BAR Router	Se0/0/0	16.65.6.18	255.255.255.252	—	Manual
Barishal	BAR Router	Se0/0/1	16.65.6.21	255.255.255.252	—	Manual

Unit	Device Name	Interface	IP Address	Subnet Mask	Default Gateway	Assignment
Barishal	DHCP - BAR	Fa0	16.65.5.2	255.255.255.128	16.65.5.1	Manual
Barishal	EMAIL - BAR	Fa0	16.65.5.3	255.255.255.128	16.65.5.1	Manual
Barishal	BAR - Printer	Fa0	16.65.5.4	255.255.255.128	16.65.5.1	Manual
Barishal	BAR - PC1	Fa0	16.65.5.10	255.255.255.128	16.65.5.1	DHCP (Local)
Barishal	BAR - PC2	Fa0	16.65.5.11	255.255.255.128	16.65.5.1	DHCP (Local)
Khulna	KHU Router	Gi0/0	16.65.5.129	255.255.255.128	—	Manual
Khulna	KHU Router	Se0/0/0	16.65.6.14	255.255.255.252	—	Manual
Khulna	KHU Router	Se0/0/1	16.65.6.17	255.255.255.252	—	Manual
Khulna	DHCP - KHU	Fa0	16.65.5.130	255.255.255.128	16.65.5.129	Manual
Khulna	EMAIL - KHU	Fa0	16.65.5.131	255.255.255.128	16.65.5.129	Manual
Khulna	KHU - Printer	Fa0	16.65.5.132	255.255.255.128	16.65.5.129	Manual
Khulna	KHU - PC1	Fa0	16.65.5.140	255.255.255.128	16.65.5.129	DHCP (Local)
Khulna	KHU - PC2	Fa0	16.65.5.141	255.255.255.128	16.65.5.129	DHCP (Local)

4. Router Running-Configurations & CLI Commands

To establish a fully functional, redundant communication network, specific running-configuration commands were implemented on all six routers. The network operates three core routing domains meeting at DHK:

1. Dynamic RIPv2 Routing Domain: Connects Dhaka, Rajshahi, and Sylhet via a shared gigabit switch segment, allowing dynamic, classless propagation of LAN prefixes with automatic summarization disabled.

2. Static Domain with Failover: Configures Khulna and Barishal using point-to-point administrative serial links, using recursive floating static routes with high administrative distances (AD 10) to bypass failed primary links.
3. Stub Network: Configures Chattogram with a single default route pointing to Dhaka.

4.1 Dhaka Hub Router (DHK) Configuration

DHK acts as the primary transit node. It executes RIPv2, redistributes static routes, and hosts four distinct DHCP pools to automatically lease addresses across the routed dynamic segment.

```
Router> enable
Router# configure terminal
Router(config)# hostname DHK
DHK(config)#
DHK(config)# ! Configure LAN Interface
DHK(config)# interface GigabitEthernet0/0
DHK(config-if)# ip address 16.65.0.1 255.255.254.0
DHK(config-if)# no shutdown
DHK(config-if)# exit
DHK(config)#
DHK(config)# ! Configure Central Switch Interface
DHK(config)# interface GigabitEthernet0/1
DHK(config-if)# ip address 16.65.6.1 255.255.255.248
DHK(config-if)# no shutdown
DHK(config-if)# exit
DHK(config)#
DHK(config)# ! Configure Serial WAN Interfaces
DHK(config)# interface Serial0/0/0
DHK(config-if)# ip address 16.65.6.9 255.255.255.252
DHK(config-if)# clock rate 64000
DHK(config-if)# no shutdown
DHK(config-if)# exit
DHK(config)#
DHK(config)# interface Serial0/0/1
DHK(config-if)# ip address 16.65.6.13 255.255.255.252
DHK(config-if)# clock rate 64000
DHK(config-if)# no shutdown
DHK(config-if)# exit
DHK(config)#
DHK(config)# interface Serial0/1/0
DHK(config-if)# ip address 16.65.6.22 255.255.255.252
DHK(config-if)# clock rate 64000
DHK(config-if)# no shutdown
DHK(config-if)# exit
DHK(config)#
DHK(config)# ! Configure DHCP Pools for Hub, Stub, and RIP LANs
DHK(config)# ip dhcp excluded-address 16.65.0.1 16.65.0.10
DHK(config)# ip dhcp excluded-address 16.65.2.1 16.65.2.10
DHK(config)# ip dhcp excluded-address 16.65.3.1 16.65.3.10
DHK(config)# ip dhcp excluded-address 16.65.4.1 16.65.4.10
DHK(config)#
DHK(config)# ip dhcp pool DHK_LAN
DHK(dhcp-config)# network 16.65.0.0 255.255.254.0
DHK(dhcp-config)# default-router 16.65.0.1
```

```

DHK(dhcp-config)# dns-server 16.65.0.2
DHK(dhcp-config)# exit
DHK(config)#
DHK(config)# ip dhcp pool CTG_LAN
DHK(dhcp-config)# network 16.65.2.0 255.255.255.0
DHK(dhcp-config)# default-router 16.65.2.1
DHK(dhcp-config)# dns-server 16.65.0.2
DHK(dhcp-config)# exit
DHK(config)#
DHK(config)# ip dhcp pool RAJ_LAN
DHK(dhcp-config)# network 16.65.3.0 255.255.255.0
DHK(dhcp-config)# default-router 16.65.3.1
DHK(dhcp-config)# dns-server 16.65.0.2
DHK(dhcp-config)# exit
DHK(config)#
DHK(config)# ip dhcp pool SYL_LAN
DHK(dhcp-config)# network 16.65.4.0 255.255.255.0
DHK(dhcp-config)# default-router 16.65.4.1
DHK(dhcp-config)# dns-server 16.65.0.2
DHK(dhcp-config)# exit
DHK(config)#
DHK(config)# ! Configure Static & Redundant Paths
DHK(config)# ip route 16.65.2.0 255.255.255.0 16.65.6.10
DHK(config)# ip route 16.65.5.0 255.255.255.128 16.65.6.21
DHK(config)# ip route 16.65.5.0 255.255.255.128 16.65.6.14 10
DHK(config)# ip route 16.65.5.128 255.255.255.128 16.65.6.14
DHK(config)# ip route 16.65.5.128 255.255.255.128 16.65.6.21 10
DHK(config)# ip route 16.65.6.16 255.255.255.252 16.65.6.14
DHK(config)# ip route 16.65.6.16 255.255.255.252 16.65.6.21 10
DHK(config)#
DHK(config)# ! Configure Dynamic RIPv2 with Metric Redistribution
DHK(config)# router rip
DHK(config-router)# version 2
DHK(config-router)# network 16.0.0.0
DHK(config-router)# redistribute static metric 5
DHK(config-router)# passive-interface Serial0/0/0
DHK(config-router)# passive-interface Serial0/0/1
DHK(config-router)# passive-interface Serial0/1/0
DHK(config-router)# no auto-summary
DHK(config-router)# end
DHK# copy running-config startup-config

```

4.2 Chattogram Stub Router (CTG) Configuration

CTG connects exclusively to DHK. It relies on a default route pointing to DHK's serial next hop, and is configured with a DHCP helper address on its LAN interface to relay client requests back to DHK.

```

Router> enable
Router# configure terminal
Router(config)# hostname CTG
CTG(config)#
CTG(config)# ! Configure LAN Interface with DHCP Helper

```

```

CTG(config)# interface GigabitEthernet0/0
CTG(config-if)# ip address 16.65.2.1 255.255.255.0
CTG(config-if)# ip helper-address 16.65.6.1
CTG(config-if)# no shutdown
CTG(config-if)# exit
CTG(config)#
CTG(config)# ! Configure WAN Serial link to DHK
CTG(config)# interface Serial0/0/0
CTG(config-if)# ip address 16.65.6.10 255.255.255.252
CTG(config-if)# no shutdown
CTG(config-if)# exit
CTG(config)#
CTG(config)# ! Configure Stub Default Route pointing to DHK
CTG(config)# ip route 0.0.0.0 0.0.0.0 16.65.6.9
CTG(config)# end
CTG# copy running-config startup-config

```

4.3 Rajshahi Board Router (RAJ) Configuration

RAJ is a direct participant in the central dynamic switch network. It exchanges routes via RIPv2 and relays dynamic client requests to DHK's DHCP scopes.

```

Router> enable
Router# configure terminal
Router(config)# hostname RAJ
RAJ(config)#
RAJ(config)# ! Configure LAN Interface with DHCP Helper
RAJ(config)# interface GigabitEthernet0/0
RAJ(config-if)# ip address 16.65.3.1 255.255.255.0
RAJ(config-if)# ip helper-address 16.65.6.1
RAJ(config-if)# no shutdown
RAJ(config-if)# exit
RAJ(config)#
RAJ(config)# ! Configure Central Switch Gigabit Interface
RAJ(config)# interface GigabitEthernet0/1
RAJ(config-if)# ip address 16.65.6.3 255.255.255.248
RAJ(config-if)# no shutdown
RAJ(config-if)# exit
RAJ(config)#
RAJ(config)# ! Configure Dynamic Routing via RIPv2
RAJ(config)# router rip
RAJ(config-router)# version 2
RAJ(config-router)# network 16.0.0.0
RAJ(config-router)# no auto-summary
RAJ(config-router)# end
RAJ# copy running-config startup-config

```

4.4 Sylhet Board Router (SYL) Configuration

SYL participates alongside RAJ and DHK in the RIPv2 dynamic exchange, and relays DHCP DISCOVER broadcasts to DHK's server.

```
Router> enable
Router# configure terminal
Router(config)# hostname SYL
SYL(config)#
SYL(config)# ! Configure LAN Interface with DHCP Helper
SYL(config)# interface GigabitEthernet0/0
SYL(config-if)# ip address 16.65.4.1 255.255.255.0
SYL(config-if)# ip helper-address 16.65.6.1
SYL(config-if)# no shutdown
SYL(config-if)# exit
SYL(config)#
SYL(config)# ! Configure Central Switch Gigabit Interface
SYL(config)# interface GigabitEthernet0/1
SYL(config-if)# ip address 16.65.6.2 255.255.255.248
SYL(config-if)# no shutdown
SYL(config-if)# exit
SYL(config)#
SYL(config)# ! Configure Dynamic Routing via RIPv2
SYL(config)# router rip
SYL(config-router)# version 2
SYL(config-router)# network 16.0.0.0
SYL(config-router)# no auto-summary
SYL(config-router)# end
SYL# copy running-config startup-config
```

4.5 Khulna Board Router (KHU) Configuration

KHU uses a static routing layout. To support robust path redundancy, each static route via DHK is mirrored with a floating static route via BAR at an Administrative Distance of 10.

```
Router> enable
Router# configure terminal
Router(config)# hostname KHU
KHU(config)#
KHU(config)# ! Configure LAN Interface
KHU(config)# interface GigabitEthernet0/0
KHU(config-if)# ip address 16.65.5.129 255.255.255.128
KHU(config-if)# no shutdown
KHU(config-if)# exit
KHU(config)#
KHU(config)# ! Configure Serial WAN link to DHK (Primary)
KHU(config)# interface Serial0/0/0
KHU(config-if)# ip address 16.65.6.14 255.255.255.252
KHU(config-if)# clock rate 64000
KHU(config-if)# no shutdown
KHU(config-if)# exit
KHU(config)#
```

```

KHU(config)# ! Configure Serial WAN link to BAR (Redundant Partner)
KHU(config)# interface Serial0/0/1
KHU(config-if)# ip address 16.65.6.17 255.255.255.252
KHU(config-if)# clock rate 64000
KHU(config-if)# no shutdown
KHU(config-if)# exit
KHU(config)#
KHU(config)# ! Primary Static Routes (Next-hop DHK: 16.65.6.13, Metric/AD: 1)
KHU(config)# ip route 16.65.0.0 255.255.254.0 16.65.6.13
KHU(config)# ip route 16.65.2.0 255.255.255.0 16.65.6.13
KHU(config)# ip route 16.65.3.0 255.255.255.0 16.65.6.13
KHU(config)# ip route 16.65.4.0 255.255.255.0 16.65.6.13
KHU(config)# ip route 16.65.6.0 255.255.255.248 16.65.6.13
KHU(config)# ip route 16.65.6.8 255.255.255.252 16.65.6.13
KHU(config)# ip route 16.65.6.20 255.255.255.252 16.65.6.13
KHU(config)# ip route 16.65.5.0 255.255.255.128 16.65.6.18
KHU(config)#
KHU(config)# ! Floating Static Backup Routes (Next-hop BAR: 16.65.6.18, AD: 10)
KHU(config)# ip route 16.65.0.0 255.255.254.0 16.65.6.18 10
KHU(config)# ip route 16.65.2.0 255.255.255.0 16.65.6.18 10
KHU(config)# ip route 16.65.3.0 255.255.255.0 16.65.6.18 10
KHU(config)# ip route 16.65.4.0 255.255.255.0 16.65.6.18 10
KHU(config)# ip route 16.65.6.0 255.255.255.248 16.65.6.18 10
KHU(config)# ip route 16.65.6.8 255.255.255.252 16.65.6.18 10
KHU(config)# ip route 16.65.6.12 255.255.255.252 16.65.6.18 10
KHU(config)# ip route 16.65.6.20 255.255.255.252 16.65.6.18 10
KHU(config)# end
KHU# copy running-config startup-config

```

4.6 Barishal Board Router (BAR) Configuration

BAR replicates the static-backup framework, providing dynamic recursive path failover. Its primary routes point directly to DHK with floating backup paths established via KHU.

```

Router> enable
Router# configure terminal
Router(config)# hostname BAR
BAR(config)#
BAR(config)# ! Configure LAN Interface
BAR(config)# interface GigabitEthernet0/0
BAR(config-if)# ip address 16.65.5.1 255.255.255.128
BAR(config-if)# no shutdown
BAR(config-if)# exit
BAR(config)#
BAR(config)# ! Configure Serial WAN link to DHK (Primary)
BAR(config)# interface Serial0/0/1
BAR(config-if)# ip address 16.65.6.21 255.255.255.252
BAR(config-if)# clock rate 64000
BAR(config-if)# no shutdown
BAR(config-if)# exit
BAR(config)#
BAR(config)# ! Configure Serial WAN link to KHU (Redundant Partner)
BAR(config)# interface Serial0/0/0

```

```

BAR(config-if)# ip address 16.65.6.18 255.255.255.252
BAR(config-if)# clock rate 64000
BAR(config-if)# no shutdown
BAR(config-if)# exit
BAR(config)#
BAR(config)# ! Primary Static Routes (Next-hop DHK: 16.65.6.22, Metric/AD: 1)
BAR(config)# ip route 16.65.0.0 255.255.254.0 16.65.6.22
BAR(config)# ip route 16.65.2.0 255.255.255.0 16.65.6.22
BAR(config)# ip route 16.65.3.0 255.255.255.0 16.65.6.22
BAR(config)# ip route 16.65.4.0 255.255.255.0 16.65.6.22
BAR(config)# ip route 16.65.6.0 255.255.255.248 16.65.6.22
BAR(config)# ip route 16.65.6.8 255.255.255.252 16.65.6.22
BAR(config)# ip route 16.65.6.12 255.255.255.252 16.65.6.22
BAR(config)# ip route 16.65.5.128 255.255.255.128 16.65.6.17
BAR(config)#
BAR(config)# ! Floating Static Backup Routes (Next-hop KHU: 16.65.6.17, AD: 10)
BAR(config)# ip route 16.65.0.0 255.255.254.0 16.65.6.17 10
BAR(config)# ip route 16.65.2.0 255.255.255.0 16.65.6.17 10
BAR(config)# ip route 16.65.3.0 255.255.255.0 16.65.6.17 10
BAR(config)# ip route 16.65.4.0 255.255.255.0 16.65.6.17 10
BAR(config)# ip route 16.65.6.0 255.255.255.248 16.65.6.17 10
BAR(config)# ip route 16.65.6.8 255.255.255.252 16.65.6.17 10
BAR(config)# ip route 16.65.6.20 255.255.255.252 16.65.6.17 10
BAR(config)# ip route 16.65.6.12 255.255.255.252 16.65.6.17 10
BAR(config)# end
BAR# copy running-config startup-config

```

5. Centralized and Distributed Services Architecture

Beyond structural connectivity, BoardNet integrates fully functional application layers. This includes centralized domain name services, tiered dynamic host configuration, board-specific e-mail, and specialized web hosting. This architecture ensures high availability, scalability, and administrative isolation.

5.1 Central DNS Configuration

All board networks point to the central, authoritative DNS server located in the Dhaka Board LAN at 16.65.0.2. To facilitate domain resolution across the entire workspace, 14 canonical A records were configured on the server, mapping hostnames to their respective static IP allocations.

Hostname	Record Type	Target IP Address	Points To / Device role
www.dhk.edu.bd	A	16.65.0.4	WEB - DHK (Dhaka Central Web Server)
www.ctg.edu.bd	A	16.65.2.3	WEB - CTG (Chattogram Web Server)

Hostname	Record Type	Target IP Address	Points To / Device role
dhk.edu.bd	A	16.65.0.3	EMAIL - DHK (Dhaka Email Server)
mail.dhk.edu.bd	A	16.65.0.3	EMAIL - DHK (Dhaka Email Server alias)
ctg.edu.bd	A	16.65.2.2	EMAIL - CTG (Chattogram Email Server)
mail.ctg.edu.bd	A	16.65.2.2	EMAIL - CTG (Chattogram Email Server alias)
raj.edu.bd	A	16.65.3.2	EMAIL - RAJ (Rajshahi Email Server)
mail.raj.edu.bd	A	16.65.3.2	EMAIL - RAJ (Rajshahi Email Server alias)
syl.edu.bd	A	16.65.4.2	EMAIL - SYL (Sylhet Email Server)
mail.syl.edu.bd	A	16.65.4.2	EMAIL - SYL (Sylhet Email Server alias)
bar.edu.bd	A	16.65.5.3	EMAIL - BAR (Barishal Email Server)
mail.bar.edu.bd	A	16.65.5.3	EMAIL - BAR (Barishal Email Server alias)
khu.edu.bd	A	16.65.5.131	EMAIL - KHU (Khulna Email Server)
mail.khu.edu.bd	A	16.65.5.131	EMAIL - KHU (Khulna Email Server alias)

5.2 DHCP IP Assignment Architecture

BoardNet combines centralized router-based DHCP and distributed local server-based DHCP to allocate client IP addresses dynamically. The Dhaka (DHK) router acts as the DHCP server for four boards, utilizing helper addresses on intermediate segments, while Khulna and Barishal use dedicated local DHCP server devices.

A. DHCP Relay Agent Mechanism: For remote LAN segments (CTG, RAJ, SYL) to receive leases from DHK, an 'ip helper-address 16.65.6.1' command is configured under their GigabitEthernet0/0 interfaces. Intermediate routers intercept DHCP broadcast DISCOVER requests and unicast them to DHK's central switch interface. DHK selects the pool by analyzing the gateway IP address (giaddr) field of the forwarded packet.

B. DHCP Pools and Excluded Address Configuration:

Pool Name	Server Host	Network Scope	Subnet Mask	Def. Gateway	Excluded / Start IP
DHK_LAN	DHK Router	16.65.0.0	255.255.254.0	16.65.0.1	16.65.0.1 - 16.65.0.10
CTG_LAN	DHK Router (Relayed)	16.65.2.0	255.255.255.0	16.65.2.1	16.65.2.1 - 16.65.2.10
RAJ_LAN	DHK Router (Relayed)	16.65.3.0	255.255.255.0	16.65.3.1	16.65.3.1 - 16.65.3.10
SYL_LAN	DHK Router (Relayed)	16.65.4.0	255.255.255.0	16.65.4.1	16.65.4.1 - 16.65.4.10
BAR_LAN	DHCP - BAR (Local)	16.65.5.0	255.255.255.128	16.65.5.1	Start IP: 16.65.5.10 (Local)
KHU_LAN	DHCP - KHU (Local)	16.65.5.128	255.255.255.128	16.65.5.129	Start IP: 16.65.5.140 (Local)

5.3 Web Server Configuration Details

Dedicated web servers are hosted exclusively on the Dhaka and Chattogram segments. These servers run Packet Tracer's integrated HTTP/HTTPS daemon to serve portals containing administrative links and results. Full reachability is achieved via route advertisement and the matching DNS records.

Server Name	IP Address	Configured Domain	Associated Board	Service Status
WEB - DHK	16.65.0.4	www.dhk.edu.bd	Dhaka Board (Central)	HTTP/HTTPS (Port 80/443) Enabled
WEB - CTG	16.65.2.3	www.ctg.edu.bd	Chattogram Board	HTTP/HTTPS (Port 80/443) Enabled

5.4 Email Server Configuration Details

Each board segment hosts a dedicated, local Email server to enable secure administrative communication and coordinate results across different regional domains. The email servers operate SMTP (TCP port 25) and POP3 (TCP port 110) protocols with DNS MX records resolved locally.

Server Name	IP Address	E-mail Domain Name	HSC Board Region	Standard User Setup
EMAIL - DHK	16.65.0.3	mail.dhk.edu.bd	Dhaka (DHK)	admin@dhk.edu.bd, controller@dhk.edu.bd
EMAIL - CTG	16.65.2.2	mail.ctg.edu.bd	Chattogram (CTG)	admin@ctg.edu.bd, staff@ctg.edu.bd
EMAIL - RAJ	16.65.3.2	mail.raj.edu.bd	Rajshahi (RAJ)	admin@raj.edu.bd, staff@raj.edu.bd
EMAIL - SYL	16.65.4.2	mail.syl.edu.bd	Sylhet (SYL)	admin@syl.edu.bd, staff@syl.edu.bd
EMAIL - BAR	16.65.5.3	mail.bar.edu.bd	Barishal (BAR)	admin@bar.edu.bd, staff@bar.edu.bd
EMAIL - KHU	16.65.5.131	mail.khu.edu.bd	Khulna (KHU)	admin@khu.edu.bd, staff@khu.edu.bd